Vertical Flight Infrastructure Data Quality Shortcomings

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Abstract

This paper represents an assessment of how vertical flight infrastructure data is accounted for, the criteria and standards used to accomplish this, what shortcomings may exist in both the data as well as the process, and what changes need to be implemented in preparing for the next generation of vertical flight transportation. The primary focus of this paper will be to review the regulations, criteria, standards, and processes that apply to aviation infrastructure data Collection, Accuracy, Resolution, Integrity, Traceability, Timeliness, Completeness, and Format and to what extent it is successfully being accomplished in the vertical flight industry. Based on these findings, this paper will strive to identify the current gaps in data accountability for vertical flight infrastructure and provide recommendations for remediating these issues. While this assessment will be specific to current and traditional helicopter infrastructure, it is intended to provide insights into identifying best practices as it applies to the Advanced Air Mobility (AAM) industry and its supporting infrastructure as well.

Introduction

Quality data is the cornerstone for basing good decisions. As it applies to aviation infrastructure these decisions include, but are not limited to, such things as site selection, site protection, preflight planning, in-flight planning, emergency response, risk analysis, asset allocation, funding distributions, business model development, community integration, community acceptance, and equity. The primary data repository in the United States for aviation infrastructure is the Federal Aviation Administrations (FAA) Airport Master Record (ARM) database. All suppliers of aviation infrastructure data pull their data from this sole source database making accuracy and accountability critical. The primary interface for users to this database is through the FAA's Airport Data and Information Portal (ADIP).[1] The FAA also promotes the open exchange of safety information in order to continuously improve aviation safety and infrastructure data is included in this information. To further this basic objective, the FAA developed the Aviation Safety Information Analysis and Sharing (ASIAS) system for this purpose.[2] The ASIAS system enables users to perform integrated queries across multiple databases, search an extensive warehouse of safety data, and display pertinent elements in an array of useful formats. However, without accurate quality data, these efforts become stymied at best, and at worst, allow for incorrect assumptions and decisions to be made by everyone involved.

For new entrance coming into the aviation market such as electric vertical takeoff and landing (eVTOL) aircraft to flourish as a true public transportation model, good policy that properly addresses data quality and accountability is critical to future success. In building for the future, we must first ensure that the foundation we are building on is in fact solid and capable. This paper will provide the reader with key data elements that the FAA and industry will need to ensure are accurately collected, tracked, regularly monitored, and properly updated to make this vision a reality. This paper will provide the reader with examples of identified challenges and gaps that need to be addressed throughout the vertical flight and aviation industries.

Oversight and Responsibility

49 U.S.C. 329(b) empowers and directs the Secretary of Transportation to collect and disseminate information on civil aeronautics.[3] Aeronautical information is required by the FAA in order to carry out agency missions such as those related to aviation flying safety, flight planning, airport engineering and federal grants analysis,

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aeronautical chart and flight information publications, and the promotion of air commerce as required by statute. The safety information collected includes, but is not limited to, the following: Airport name, associated city, airport owner and airport manager, airport latitude, longitude, elevation, runway description, services available, runway approach light systems, communications frequency, airport use, number of operations and based aircraft, obstruction data, and pertinent general remarks.[4]

However, while the above criteria apply to public use facilities the same cannot be said for private use facilities. For private use facilities the responsibility resides with the owner of the facility to ensure the data on file is up to date and correct. As of 2004 the FAA heliport design guide states that for Prior Permission Required (PPR) heliports, which includes all private use heliport, that:

> *"The heliport owner and operator should ensure that all pilots are thoroughly knowledgeable with the heliport (including such features as approach/ departure path characteristics, preferred heading, facility limitations, lighting, obstacles in the area, size of the facility, etc.)."*

While there are a handful of states, such as California, who do facilitate the collection and validation of data for heliports located specifically at hospitals, they are the exception.

Forms and Records

The primary forms used by the FAA for gathering and accounting for data regarding aviation infrastructure are listed in Table-1:

Table-1 FAA Forms and Records

FAA Form	Title
7460	Notice Of Proposed Construction Or Alteration
7480	Notice For Construction, Alteration And Deactivation Of Airports
5010 (1, 2, 3, 4, 5)	Airport Master Record

**Note: There are a number of different versions of each form.*

Form 7480-1 – Notice for Construction, Alteration and Deactivation of Airports (and Heliports). Per Title 14 Code of Federal Regulations Part 157, to notify the FAA at least 90 days before construction, alteration, activation, deactivation, or change to the status or use of a civil or joint-use (civil/military) airport.

The Airport Master Record (AMR) refers to all four FAA forms used in the Airport Safety Data Program – 5010-1; 5010-2; 5010-3 and 5010-5. Forms 5010-1 & 5010-2 are for existing airports/heliports and contain aeronautical data on existing landing areas. Form 5010-1 is for civil public-use airports/heliports, joint military and public-use airports/heliports and private-use military airports/heliports. Form 5010-2 is for civil private-use airports/heliports. Form 5010-3 and 5010-5 are "blank" forms used to report aeronautical data on a newly proposed or newly constructed airport/heliport. Form 5010-3 is used to provide the initial physical inspection report on a newly constructed or reported civil public-use airport/heliport. Form 5010-5 is used to provide an initial report on a newly constructed or newly reported civil private-use airport/heliport.[5]

Heliport Supplemental Data

With the development and publication of the FAA ADIP system a stop gap measure was instituted to assist in collecting more heliport centric data. This involved the inclusion of a spread sheet for each heliport that contained heliport specific data. However, given the lack of funding, resources, and staffing within the FAA Airport Division, along with higher priority projects, the spreadsheets were populated with original data from the AMR database. While a good effort to move toward better vertical flight data accountability, these spreadsheets have never been validated.

Accuracy Implications

The information contained within the FAA AMR Database is used extensively by other segments of the civil and military aeronautical communities and is the agency's sole source for the collection and dissemination of airport safety data used in aeronautical and flight information publications as well as GPS databases. Users of the FAA data include but are not limited to Jeppesen, ForeFlight, Aircraft Owners, Pilots, Aviation Associations, State Agencies, State Airport Directories, Aeronautical Publications, System Planners, and other government agencies such as the National Oceanic & Atmospheric Administration and the National Geospatial-Intelligence Agency. What this then implies is that if the information is incorrect or missing in the FAA AMR Database it will be missing or incorrect in all the aforementioned systems and databases.

A good example of the criticality of accurate information can be found in the advertisement material for a portable aviation GPS, *"aera 796",* offered by Garmin, states the following in its marketing material:

> **"…**while a private airport and heliport database greatly expands your emergency "nearest" options."[6]

The key element that must be understood when it comes to private use facilities in the U.S. when compared to public use facilities is that the FAA is not required to, and does not validate, the information in the database for these facilities. There are no Federal requirements for a private use heliport to ever be reinspected, audited, or reevaluated after it is activated unless a major change in the facility has occurred. The new FAA ADIP database contains a data field for private use heliports identified as "Last Owner Information Date" (Item A113 on FAA Form 5010). This indicates the last date an information response was completed by facility owner or manager.[7] Previous guidance for private use facilities indicated to enter the date that the proponent completed the Form 5010-5.[8] A data review showed 691 individual heliports with blank data fields for A113 while 962 were found to be over 20 years old.

Heliport Statistics

The following statistics for the United States are based on current data (March 2024) downloaded from the FAA AMR/ADIP system. Table-2 provides a breakdown by ownership and Table-3 provides a breakdown by use case.

Table-2 U.S. Heliport Ownership Breakdown

U.S. Heliport Ownership	Number
Public Owned	635
Private Owned	5,454
U.S. Coast Guard	4
U.S. Air Force	6
U.S. Navy	14
U.S. Army	72
Total U.S. Heliports	6,185

Table-3 U.S. Heliport Use Case Breakdown

U.S. Heliport Use Case		Number
Public-Use		55
Private-Use		6,130
*Medical-Use**		*2,849*

**Note: Medical-Use is a subcategory of public and private use. There are 3 public medical use heliports and 2,846 private medical use heliports listed in the ADIP system.*

Medical Use vs. Hospital

The FAA's AMR database currently accounts for 2,849 "Medical-Use" heliports in the U.S. However, due to poor taxonomy of terminology this number does not represent the total number of "Hospital Heliports" that exist in the U.S. In numerous cases "Medical-Use" heliports represent heliport locations that are collocated at Fire Stations, Police Posts, EMS Bases, and other sites such as federal and state parks which are not connected with a healthcare facility. In 2024 the American Hospital Association reported a total of 6,120 hospitals in operation in the U.S., of which 5,129 were categorized as Community Hospitals, and 207 as Federal Government Hospitals. Table-4 provides a breakdown of Hospitals in the U.S.[9]

Table-4 U.S. Hospital Breakdown

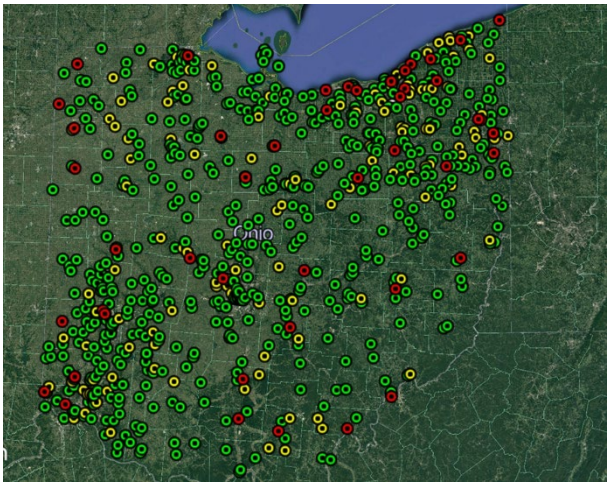
U.S. Hospitals	Number
Community Hospitals	5,129
Federal Government Hospitals	207
Nonfederal Psychiatric Hospitals	659
Other Hospitals	125
Total:	6,120

In 2019, NASA determined that there was an estimated underreporting of U.S. hospital heliport of upwards of 2,000 in the FAA's AMR database system.[10]

To compound this issue, several of the heliport facilities identified as a Healthcare facility in the data field label "Name" in the AMR using the terms: Hospital, Medical Center, Medical Park, Health, Healthcare, Clinic, Memorial, Care, Emergency… are not identified as Medical Use facilities. This accounted for 332 healthcare facilities that were not identified as Medical Use.

In a 2018 data accountability study conducted by HeliExperts International LLC in collaboration with the U.S. Helicopter Safety Team (USHST) five states were reviewed. Figure-1 illustrates the results of the findings for the state of Ohio. At the time, there were 44 active hospital heliports, shown in red, in operation that were not accounted for in the FAA AMR.[11]

Figure-1 Ohio Infrastructure Accountability Study



Type	No.
Airports	428
Heliports	231
Seaplane Bases	3
Gliderports	2
Balloonports	1
Total	665

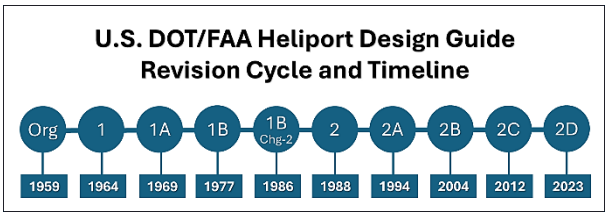
Verified	537
Inaccurate or Missing	128
Unlisted/Unknown	44

Advisory Circular Evolution

Heliport standards as we know them today were developed in accordance with the directive found in the Federal Aviation Act of 1958 that created the Federal Aviation Agency (later the Federal Aviation Administration or the FAA) and abolished its predecessor, the Civil Aeronautics Administration (CAA). Section 103 of the Act states in part, *"In the exercise and performance of his power and duties under this Act, the Secretary of Transportation shall consider the following, among other things, as being in the public interest: (a) The regulation of air commerce in such manner as to best promote its development and safety and fulfill the requirements of defense; (b) The promotion, encouragement, and development of civil aeronautics . . .*"[12]

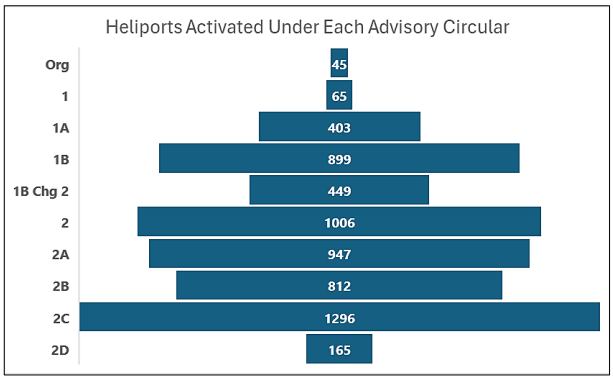
In the United States, heliport design standards are published in U.S. DOT/FAA Advisory Circular AC 150/5390. The first version of this standard was published in December of 1959. Since that date, there have been nine revisions made to this standard with the most recent one being revision "2D" published in January 2023. Table-5 shows the timeline for the revision cycle of the advisory circular. Over the years the terminology used to describe the three primary elements of a heliport that we know today as the Touchdown and Liftoff (TLOF) area, Final Approach and Takeoff (FATO) area, and the Safety Area have changed. See Table 5.

Table-5 Advisory Circular Revision Timeline



A key element regarding the application of heliport design standards is that a heliport is expected to meet the standard under which it was first designed and activated. See Table-6. If the heliport were developed in 1965, and there have never been any major changes to the heliport, the standard that must be used when evaluating that heliport is the one that was in place at the time, which in this case would have been the one published in 1964.

Table-6 Heliport Accountability by Circular Date

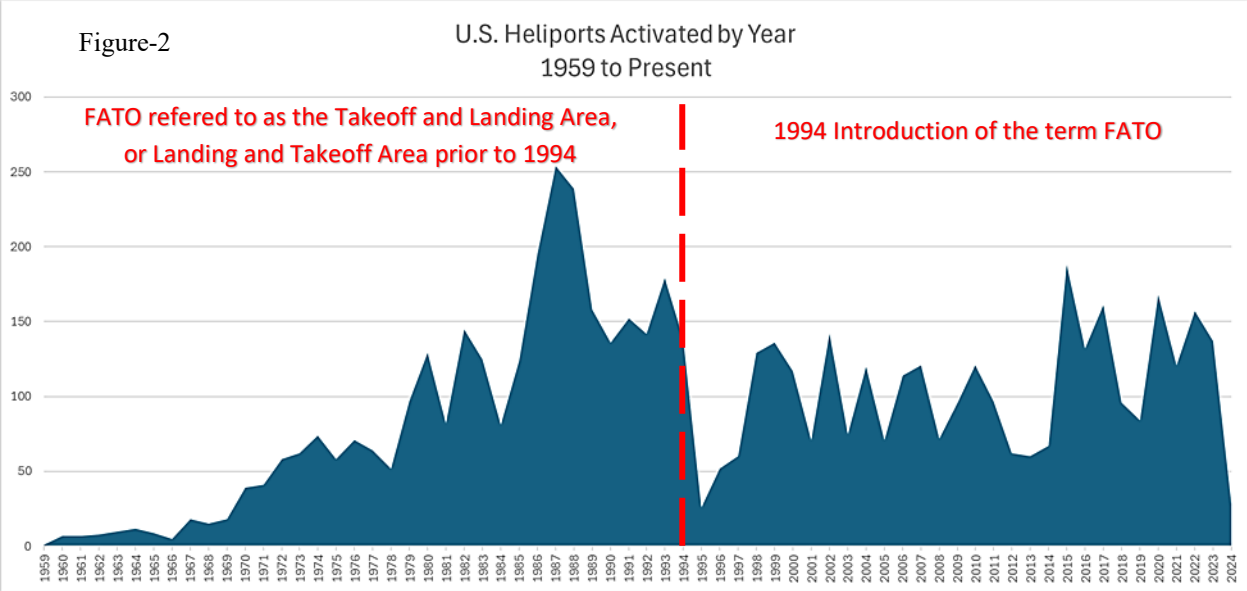


What we now refer to as the Touchdown and Liftoff (TLOF) Area was first called the Touchdown Area in the original 1959 standard then later the Touchdown Pad in the 1969, 1977, and 1986 standard. However, in 1988 the FAA changed the Touchdown Area to Final Approach and Takeoff Area (FATO). This was then later changed to Touchdown and Liftoff Area in 1994 which it has remained ever since.

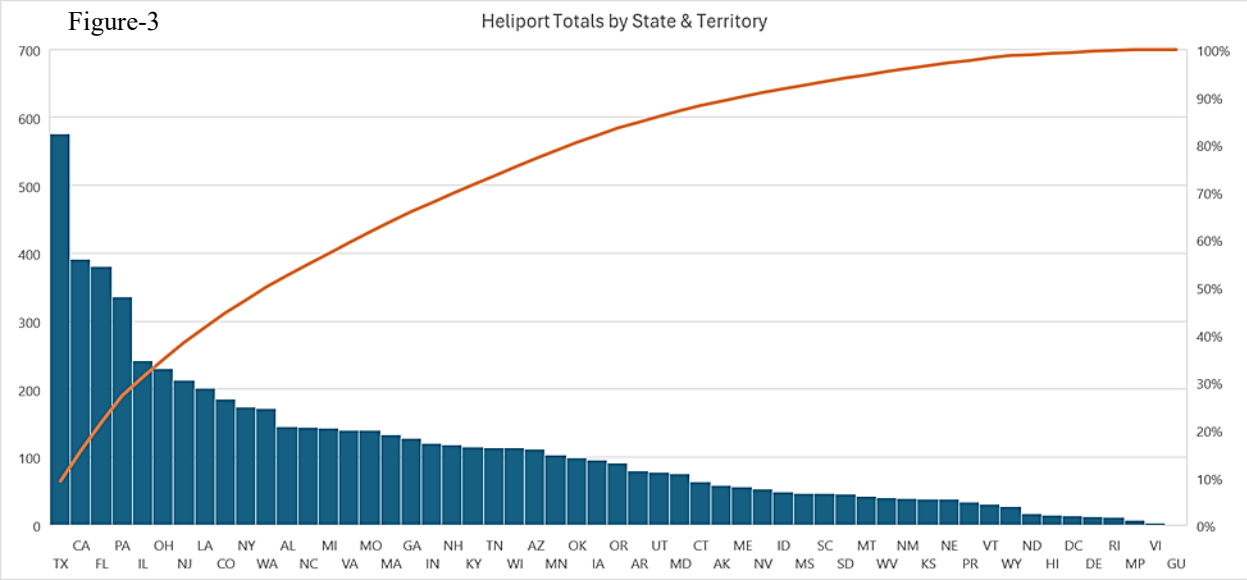
What we now refer to as the Final Approach and Takeoff (FATO) area was first called the Takeoff and Landing Area in the original 1959 standard. This was then changed to Landing and Takeoff Area in 1964 and 1969 but was changed back to Takeoff and Landing Area in 1977, 1986, and 1988. It is in 1994 that we first see the reference to the Final Approach and Takeoff (FATO) as it is currently understood.

What we now understand to be the Safety Area was originally called the Peripheral Area from 1959 to 1986. However, there is no mention of either a Peripheral Area or a Safety Area in the 1988 revision. The term Safety Area, as we know it today, first appeared in the 1994 revision.

The total number of heliports in existence prior to 1994 impacted by these changes is approximately 2,879. The impact of these changes and variations in terminology as used in successive circulars is unknown and unaccounted for in the FAA AMR database. The dimensions of a heliport dictate the size of the helicopter it can safely support. However, at the present time much of the dimensional data for heliports in the AMR is incorrect.



In Figure-2 we can see the number of heliports activated in the United States per year since 1959. There are 12 heliports on record as having existed prior to 1959 with the earliest one having an activation date of 1927.



In Figure-3 we see the distribution of heliports by state and territory. Territories represented include Guam (GU), N Mariana Islands (MP), Puerto Rico (PR), and the Virgin Islands (VI).

It should be pointed out that the FAA AMR does indicate that there were 12 heliports in existence prior to 1959. Given that the accuracy of this data is next to impossible to validate, these 12 heliport were left out of the scope of this study. If the data were correct, all of these facilities would have been established prior to the first heliport design standard being published.

Heliport Airspace

The imaginary surfaces that make up a heliport's airspace for the purposes of conducting operations under Visual Flight Rules (VFR) and maintaining hazard free airspace can be found in the Code of Federal Regulations. While the heliport design guide that first mentions FAA Part 77, Safe, Efficient Use, and Preservation of the Navigable Airspace, where these surfaces are defined, was in version "1" (1964), it was not until 1977 in version "1B" that these surfaces were defined. The following table provides the regulatory definitions of the three imaginary surfaces described.[13] See Table-7.

Table-7 Heliport Imaginary Surfaces

SURFACE(S)	DEFINITION
Primary	The area of the primary surface coincides in size and shape with the designated take-off and landing area. This surface is a horizontal plane at the elevation of the established heliport elevation.
Approach	The approach surface begins at each end of the heliport primary surface with the same width as the primary surface and extends outward and upward for a horizontal distance of 4,000 feet where its width is 500 feet. The slope of the approach surface is 8 to 1 for civil heliports and 10 to 1 for military heliports.
Transition	These surfaces extend outward and upward from the lateral boundaries of the primary surface and from the approach surfaces at a slope of 2 to 1 for a distance of 250 feet measured horizontally from the centerline of the primary and approach surfaces.

These can also be found referenced in the current FAA Heliport Design Guide, version "2D" (2023).

The heliport design guide allows for either a straight approach surface or a curved approach surface to be used when developing a heliport site location. See Figures 4 and 5.

Figure-4 Heliport Imaginary Surfaces (Straight)

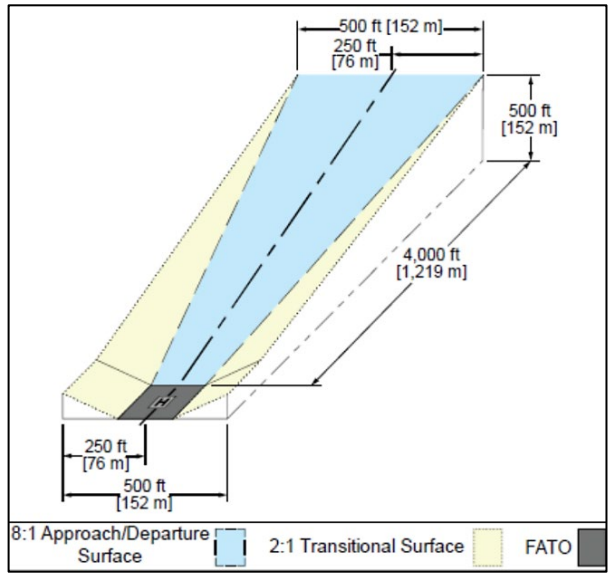
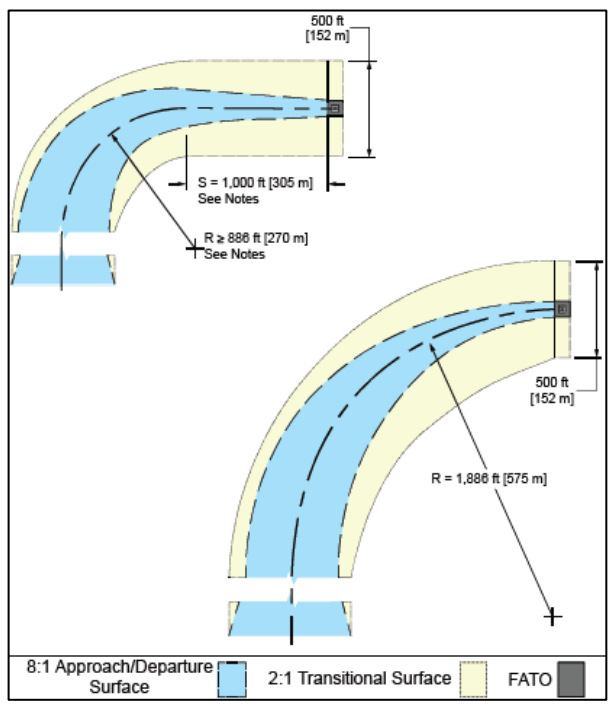


Figure-5 Heliport Imaginary Surfaces (Curved)



While FAA Form 7480 does have a data field to record a proposed heliport's approach/departure paths, FAA Form 5010 and the AMR do not have these same fields to capture this data. Additionally, the optional curved approach/departure path, which has been an option since 1959, is not captured in any data fields on either the 7480 or 5010 and is not provided for in the AMR.

For these reasons, while it is a requirement to demonstrate to the FAA that a heliport's approach surface, straight or curved, is clear of all obstructions, the data is not captured in the AMR and hence is lost to time. Figure-6 shows the information in the AMR as it is related to airports which includes the magnetic heading of the runway and hence the approach/departure path to the runway. Figure-7 shows the same data fields for heliports where only an H is used for the identifier and there are no magnetic headings included for the approach/departure paths.

Figure-6 Airport Runway Id Data

Loc Id	Runway Id	Length	Width
1M3	01/19	2700	100
14J	01/19	3050	75
GZH	01/19	5005	150
CQF	01/19	6604	100
M95	01/19	5009	80
4R3	01/19	5003	80
3M8	01/19	5144	80
GEU	01/19	7150	100
40G	01/19	4199	45
L25	01/19	2900	110

Figure-7 Heliport Id Data

Loc Id	Runway Id	Length	Width
MOB	H1	100	100
CHD	H1	100	100
FHU	H1	40	40
GYR	H1	64	64
OLS	H1	97	97
PAN	H1	50	50
SAD	H1	72	72
SEZ	H1	50	50
BIH	H1	40	40
CRQ	H1	40	50

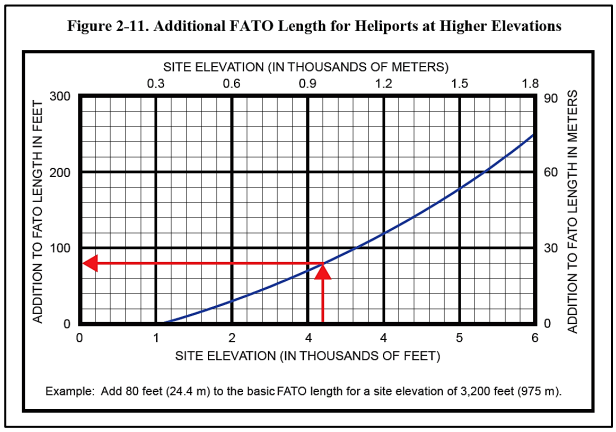
Note that in Figure-7, the AMR data does not specify what the heliports TLOF, FATO or Safety Area are, but rather provides Runway Length and Width.

Identified as one of the biggest challenges to private use heliports in the U.S. is the ability of the owner to protect the supporting airspace. The FAA stipulates in Part 77 that only public heliports and public airports, airports operated by a federal agency or the Department of Defense (DOD), and private airports and heliports with at least one FAA-approved instrument approach procedure can be protected by the FAA. Protection of Private Use VFR heliports and their airspace is the sole responsibility of the owner. The fact that the FAA does not record the approach/departure airspace in the AMR, and FAA Form 5010 has no data fields for it make it that much more difficult for owners to adequately implement zoning to protect their heliport's airspace.

Performance

While the FAA has published research related to the development of a heliport classification system based on helicopter performance in 1981, and again in 1991, to date there are no performance categories for heliports. [13, 14] There is however a stipulation that for heliports above 1,000 feet in Elevation or greater it is recommended that the FATO size be increased. According to the FAA Heliport Design Guide a longer FATO provides an increased safety margin and greater operational flexibility. See Figure-8.

Figure-8 FATO Length & Altitude Criteria

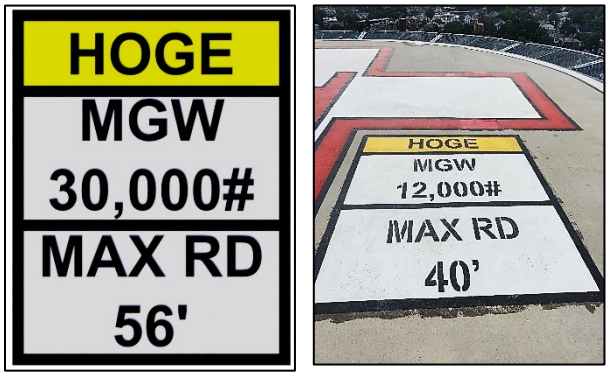


However, while this may provide some level of safety for certain types of aircraft and operations, nowhere in the AMR or on FAA Forms 7480 or 5010 is this information captured or reflected.

In leu of increasing the size of the FATO for higher elevation heliports, or heliports on rooftops, the heliport design guide provides owners with an option to restrict the heliport to aircraft that are capable of Hover-Out-Of-Ground Effect (HOGE) capability. This restriction has

been integrated by some heliport designers, who have included an additional space at the top of the heliport information box to include the letters HOGE on a yellow background. Figure-9 illustrates this type of marking and its use at a hospital heliport.

Figure-9 Heliport HOGE Marking



However, nowhere in the AMR or on FAA Form 7480 or FAA Form 5010 is this information captured or reflected.

Location Accuracy

Identifying infrastructure using an accurate set of coordinates, in this case latitude and longitude, allows for an accurate depiction of that infrastructure in the National Airspace System (NAS). If the location coordinates are incorrect this can then have a negative impact on those who may be operating in the vicinity of the location, such as UAS and drone operators.

One example of how far off a heliport's coordinates could be is the heliport TN60 in Nashville, Tennessee. AMR/ADIP data shows that this heliport was activated in May of 1979 and is located at 5700 Knob Road, Nashville, TN with the coordinates 36° 8' 28.21" N / 87° 51' 43.098" W. However, when these coordinates are plotted, they show up 55.95 statute miles due west of the actual location of the heliport. The actual coordinates for TN60 should be 36° 8'27.39"N / 86°51'41.35"W.

While this heliport was activated in 1979, which would have been under heliport design guide "1B", since it is a private use facility the FAA has no legal oversight or responsibility for its information. Since it is only a recommendation that the data be updated and not a requirement it has existed with the incorrect coordinates for over 45 years. Figure-10 shows where the heliport actually is located while Figure-11 shows where the AMR/ADIP data says it is. Figure-12 shows the total difference in statue mile between the two points.

Figure-10 TN60 Actual Location

Figure-11 TN60 Recorded Location

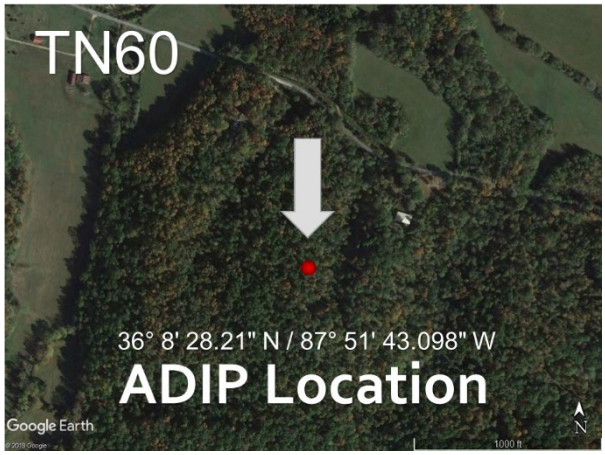
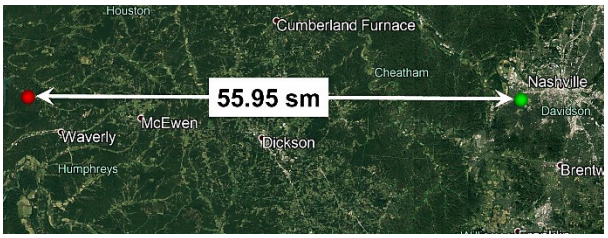


Figure-12 TN60 Coordinates Error



While this may represent an extreme case, it should be pointed out that the criteria used to identify and classify obstructions and hazards at a heliport only uses a radius of 5,000'. That of course is only if it is a public use heliport or a private use heliport with an instrument approach procedure associated with it. If it is a private

use VFR heliport there is no protection afforded the airspace and if that airspace is not where it is supposed to be it makes the situation even worse.

Airspace Determinations

While the AMR/ADIP is designed to track the airspace determinations rendered by the FAA for any given facility, there were a number of shortcomings identified in this system. The FAA will conduct an aeronautical study of an airport proposal and, after consultations with interested persons, as appropriate, issue a determination to the proponent and advise those concerned of the FAA determination.[16] Table-8 provides the terms and definitions found in the Code of Federal Regulations while, for comparison, Table-9 provides the terms and definitions used by FAA Safety Inspectors.[17]

Table-8 CFR Airport Determinations

Term	Definition
No Objection	(No Definition)
Conditional	A conditional determination will identify the objectionable aspects of a project or action and specify the conditions which must be met and sustained to preclude an objectionable determination.
Objectionable	An objectionable determination will specify the FAA's reasons for issuing such a determination.

Two specific items to point out in the above definitions from the Code of Federal Regulations. First, in the definition for Conditional, the word "MUST" is used, in that it identifies a condition that must be met and sustained to preclude an objectionable determination from being rendered. It should be pointed out that there is no system in place within the AMR/ADIP that allows for this type of check and balance to occur for private use facilities. If a Conditional determination is rendered, no one within the FAA is evaluating the site to make sure the proponent has addressed any of conditions identified. Second, while there is a definition for the term Objectionable listed, it is not a true definition as to what Objectionable actually means.

Table-9 FAA Dynamic Regulatory System (DRS) Airport Determinations

Term	Definition
No Objection	Use when there is no operational safety concern.
No Objection With Provision	Use when there is no operational safety concern (the proposal is generally acceptable to FS) but the inspector recommends specific changes or operational considerations.
Objection	Use when a proposal is clearly unacceptable due to operational safety concerns.

In comparison to the terms and definitions listed in the Code of Federal Regulations the ones utilized in the FAA Dynamic Regulatory System 8900.1 provide an interesting yet confusing contrast. Of particular interest is the fact that the FAA defines the term Objectionable as *"…clearly unacceptable due to operational safety"* however this unacceptability due to safety concerns is not conveyed in the definition provided by the CFR. The DRS evaluation standard goes on to state that:

> **"The designated Flight Standards representative heliport evaluator determines if the owner/operator can conduct safe helicopter operations at the proposed location without undue risk to the operator, or to persons and property on the ground."**

However, this is not entirely an accurate statement. While the Flight Standards representative does conduct the onsite evaluation, it is the respective Airport District Office (ADO) representative who actually writes the airspace determination letter. Situations have occurred where the Flight Standards representative has rendered an Objectionable determination, but the ADO has downgraded it to a Conditional determination. In these cases, the determination was justified in that the site evaluated was private use.

For comparison, we need to consider the three different airspace determination listed on the FAA Obstruction Evaluation / Airport Airspace Analysis (OE/AAA) website, which are used for determining the impact of structures on air navigation.[18] See Table-10.

Table-10 FAA OE/AAA

Term	Definition
Determination of No Hazard	The subject construction did not exceed obstruction standards and marking/lighting is not required.
Determination of No Hazard with Conditions	The proposed / construction alteration would be acceptable contingent upon implementing mitigating measures such as the marking and lighting of the structure.
Determination of Hazard	The proposed construction / alteration is determined to be a hazard to air navigation.

All of these differing terms and definitions must then be compared to the FAA's Procedures for Handling Airspace Matters, Order JO 7400.2P, where the FAA provides guidance for determining when an Adverse Effect exists.[19] This document specifically states that:

> **"A structure would have an adverse effect upon VFR air navigation if it penetrates the approach surface slope of any runway."**

A Heliport's 8:1 approach surface is identified as one of the surfaces to be considered in this evaluation. This document goes on to state that for a Heliport Landing/Takeoff Area:

> **"ANY structure would have an adverse effect if it would exceed ANY of the heliport imaginary surfaces."**

While the FAA goes into great detail regarding what constitutes and obstruction and when an obstruction is classified as a hazard which creates an "Adverse Effect", the majority of this criteria is not applied to private use VFR heliports, e.g., 99% of all heliports in the United States.

A good example of these shortcomings can be found in a helicopter accident that occurred on October 16, 2021, at the Tucson Medical Center Heliport (AZ37) in Tucson, Arizona. In the NTSB accident investigation report, under Airport Information, the NTSB listed the Runway Length and Width of the Heliport as being 100' X 100'. See figure 13.

Figure-13 NTSB Accident Report

Airport Information

Airport:	TUCSON MEDICAL CENTER AZ37	Runway Surface Type:	Asphalt
Airport Elevation:	2485 ft msl	Runway Surface Condition:	Dry
Runway Used:	H1	IFR Approach:	None
Runway Length/Width:	100 ft / 100 ft	VFR Approach/Landing:	Full stop

These same dimensions are reflected in the FAA AMR/ADIP system as well as the FAA ADIP Supplemental Heliport Data indicating that the TLOF at this heliport is 100' X 100' and ashplant and that there were NO Close-In Obstructions. See Figures 14 & 15.

Figure-14 ADIP Information

Helipad - H1

Dimensions:	100 ft. x 100 ft.	Surface Type/Condition:	ASPH
Latitude:	32° 15' 5.87" N	Treatment:	
Longitude:	110° 52' 45.59" W	Edge Lights Intensity:	PERI

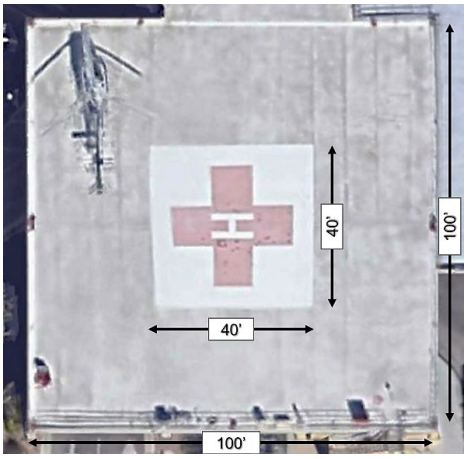
= Remark.

Figure-15 ADIP Supplemental Heliport Data

G	H	I	J	K
TLOF Length	TLOF Width	Surface Type	Approach Lights	Close-In Obstruction
100	100	ASPH	RLLS	N

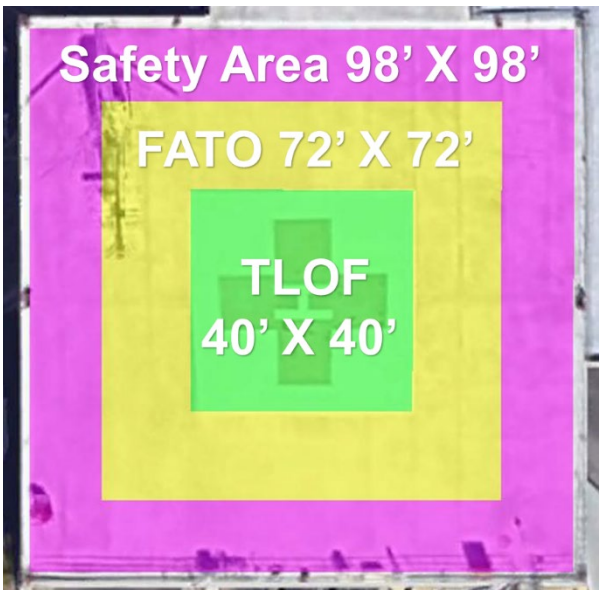
However, upon closer evaluation of the actual site, the heliport in question was found to have a 40' X 40' TLOF which resided on a concrete parking garage deck that happened to measure 100' X 100'. See Figure 16.

Figure-16 Satellite photo of heliport.



Based on an activation date of February 1990, the heliport design guide in effect at the time would have been AC 150/5390-2, which was published in 1988. Keep in mind this was prior to the introduction of the term FATO which was introduced in 1994 and the 1988 heliport guide did not reference either a Peripheral Area or a Safety Area. Figure-17 illustrates what the TLOF, FATO and Safety area would have looked like using the 1994 heliport design guidelines criteria.

Figure-17 Satellite photo of heliport.



**Dimensions based on 1994 standards.*

Additionally, the AZ37 heliport incorporates a 4' handrail around the perimeter of the 100' X 100' concrete surface, which was found to penetrate all of the 8:1 approach surfaces of the heliport. See Figures 18 & 19.

Figure-18 Handrail Dimensions and Placement

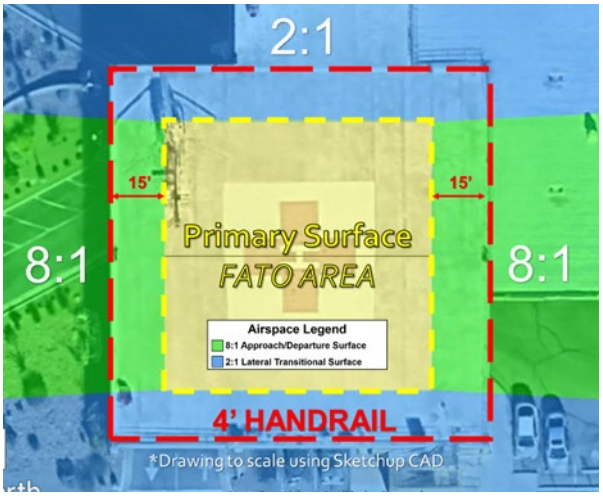
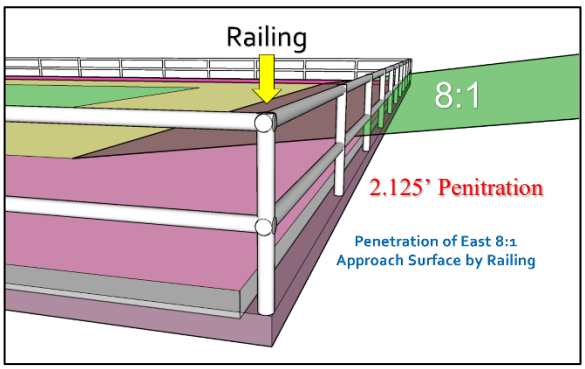


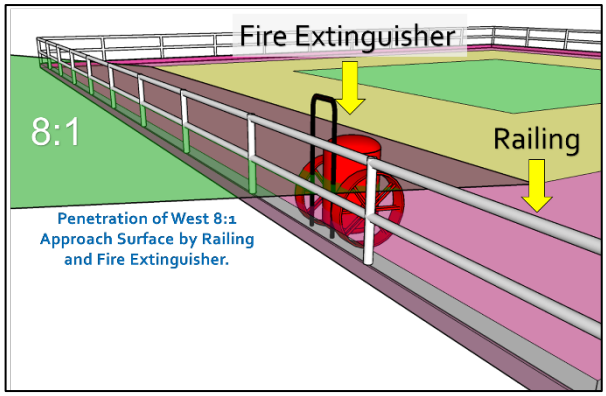
Figure-19 Handrail Penetration of 8:1 Surface.



The Airspace Determination on record for AZ37 indicates a "Conditional" determination was rendered by the FAA ADO. Beyond the Conditional determination, one website, SkyVector, included the following details, "Provided VFR/PVT USE".[20] By the FAA's own policy this site should have been classified as Objectionable due to the handrails penetration of the 8:1 approach surface. However, since it was Private Use, the Objectionable determination was likely downgraded too Conditional by the ADO in spite of what the Flight Standards inspection's report indicated.

The accident at the heliport was ultimately caused by the pilot striking a 5' tall fire extinguisher located inside the safety area that was located next to the handrail and also penetrated the 8:1 approach surface. See Figure 20.

Figure-20 Fire Extinguisher Penetration of 8:1 Surface.



In their final report the National Transportation Safety Board (NTSB) determined that the probable cause(s) of this accident to be:

"The pilot's failure to maintain clearance from the ground equipment during landing."

It should be pointed out that currently no one within the National Transportation Safety Board (NTSB) receives any training in vertical flight infrastructure standards, criteria application, or heliport accident investigation. In the accident report "WPR22LA018" published by the NTSB, there was no mention of the following:

- The fact that the heliport did not meet FAA standards.
- The information on file for the heliport in the FAA AMR/ADIP system was incorrect.
- That the 8:1 approach surface was in fact penetrated by the handrail which in turn created an unsafe high-risk condition to exist.
- That the fire extinguisher struck by the pilot on landing should have never been located on the heliport and did penetrate the 8:1 approach surface.
- That it was a frequent practice for the certificate holder to have pilots land two and sometimes three helicopters within the 100' X 100' confines of this location on a routine bases since its activation. See Figure 21.

Figure-21 Multiple helicopters on AZ37 Heliport.

All heliport design guides stipulate that a heliport is designed to support only one helicopter within the designated TLOF, FATO, and Safety Area.

At the end of the day the pilot was the one who was held responsible for the accident. Given that the above issues were never identified or addressed no action were taken to improve the site and there were no lessons learned for which the rest of the aviation industry could benefit from.

Conclusions

A fundamental overhaul of how vertical flight infrastructure, both public-use and private-use, is evaluated, documented, and tracked in the United States is long overdue. High risk and unsafe conditions exist at vertical flight infrastructure today for which pilots are required to contend with on a daily basis and when an accident occurs the pilots are the ones held responsible rather than the infrastructure owners or aircraft certificate holders.

Accident investigators desperately need the appropriate training that provides them with the tools necessary to properly evaluate accidents that occur at vertical flight infrastructure so that infrastructure can be improved rather than allowed to continue to exist with unsafe and high-risk conditions.

The information contained within the FAA AMR for vertical flight infrastructure is in desperate need of validation and updating. While the ADIP systems that the FAA has developed provides an excellent platform for the collecting, updating, and data mining of information of infrastructure in the National Airspace System, the program needs additional staffing and funding to carry out the necessary effort to make the system effective and impactful.

Individuals and organizations looking to utilize existing heliports to support Advanced Air Mobility (AAM) operations must first verify that a heliport meets or is capable of meeting current standards. A change over to support different and unique aircraft with different performance capabilities will more than likely be considered a major change by the FAA.

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